

BG770A-GL&BG95xA-GL

FTP(S) Application Note

LPWA Module Series

Version: 1.0

Date: 2022-03-31

Status: Released



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About the Document

Revision History

Version	Date	Author	Description
-	2021-01-29	Jelena JOVANOVIC	Creation of the document
1.0	2022-03-31	Jelena JOVANOVIC	First official release

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1 Introduction

Quectel BG770A-GL, BG950A-GL, and BG951A-GL modules provide FTP(S) application to FTP(S) server. This document is a reference guide to all AT commands defined for FTP(S).

1.1. Special Mark

Table 1: Special Mark

Mark	Definition
*	Unless otherwise specified, when an asterisk (*) is used after a function, feature, interface, pin name, AT command, or argument, it indicates that the function, feature, interface, pin, AT command, or argument is under development and currently not supported; and the asterisk (*) after a model indicates that the sample of such a model is currently unavailable.

1.2. The Process of Using FTP(S) AT Commands

As the modules support FTP(S) protocol, the file and directory on FTP(S) server can be operated via FTP(S) AT commands. The general process is as follows:

Step 1: Configure and activate a PDP context.

- 1) Configure <APN>, <username>, <password> and other parameters of a PDP context with **AT+QICSGP**.
- 2) Activate the PDP context with **AT+QIACT**.
- 3) Configure the PDP context ID for FTP(S) with **AT+QFTPCFG="contextid",<contextID>**.

Step 2: Configure user account and FTP(S) server.

- 1) Configure the account information with **AT+QFTPCFG="account",<username>,<password>**.
- 2) Configure the type of transferred data with **AT+QFTPCFG="filetype",<file_type>**. Either binary data or ASCII data can be transferred between FTP(S) server and client.
- 3) Configure the transfer mode with **AT+QFTPCFG="transmode",<transmode>**. The transfer

mode means either the FTP(S) server or the client listens on a port for a data connection. **AT+QFTPCFG="transmode",1** must be set for FTPS operations, because FTPS currently does not support active mode.

- 4) Configure the response timeout with **AT+QFTPCFG="rsptimeout",<timeout>**.
- 5) If the module is an FTPS client, then the following configurations are needed:
 - a) Execute **AT+QFTPCFG="ssltype",1**.
 - b) Execute **AT+QFTPCFG="sslctxid",<SSL_ctx_ID>** to select a **<SSL_ctx_ID>**.
 - c) Execute **AT+QSSLCFG** to configure the selected **<SSL_ctx_ID>**.

Step 3: Login to FTP(S) server.

Login to FTP(S) server with **AT+QFTPOPEN=<hostname>,<port>**. If **+QFTPOPEN: 0,0** URC is returned, it indicates that the operation is successful. Port numbers of FTPS and FTP servers are different. FTPS server port number depends on the server provider, and it is usually 990.

Step 4: File operation.

- 1) Set the current directory with **AT+QFTPCWD**.
- 2) Upload a file to FTP(S) server.
 - a) Solution 1: Upload a file to UFS with **AT+QFUPL**, then upload the file to FTP(S) server with **AT+QFTPPUT**. After uploading the file to FTP(S) server successfully, delete the file with **AT+QFDEL**.
 - b) Solution 2: Upload a file to FTP(S) server via COM port with **AT+QFTPPUT**, then the module enters data mode. **+++** can be inputted to finish the file uploading process.
- 3) Download a file from FTP(S) server with **AT+QFTPGET**. The file can be outputted to COM port or saved to UFS. If the file is outputted to COM port, the module enters data mode.
- 4) Get the length of data transferred between FTP(S) server and client with **AT+QFTPLEN**.
- 5) Get the size of the file on FTP(S) server with **AT+QFTPSIZE**.
- 6) Rename a file on FTP(S) server with **AT+QFTPRENAME**.
- 7) Delete a file on FTP(S) server with **AT+QFTPDEL**.
- 8) Get the file modification time on FTP(S) server with **AT+QFTPMDTM**.

Step 5: Directory operation on FTP(S) server.

- 1) Set the current directory with **AT+QFTPCWD**.
- 2) Create a directory with **AT+QFTPMKDIR**.
- 3) List the content of a directory with **AT+QFTPLIST**.
- 4) List file names of a directory with **AT+QFTPNLST**.
- 5) List standardized file and directory information with **AT+QFTPMLSD**.
- 6) Rename a directory with **AT+QFTPRENAME**.
- 7) Delete a directory with **AT+QFTPRMDIR**.

Step 6: Close the connection with FTP(S) server.

Close the connection with FTP(S) server with **AT+QFTPCLOSE**. If **+QFTPCLOSE: 0,0** URC is reported, it indicates that the operation is successful. **Step 3** to **Step 6** can be repeated.

Step 7: Deactivate PDP context.

Deactivate a PDP context with **AT+QIDEACT=<contextID>**.

NOTE

1. See **document [1]** for more information on **AT+QICSGP**, **AT+QIACT** and **AT+QIDEACT**.
2. See **document [2]** for more information on **AT+QSSLCFG**.
3. See **document [3]** for more information on **AT+QFUPL** and **AT+QFDEL**.

1.3. Description of Data Mode

The COM port of the modules has two working modes: AT command mode and data mode. In the AT command mode, the data inputted via the COM port are treated as AT commands, while in the data mode, they are treated as data.

● Exit Data Mode

Input **+++** or pull the MAIN_DTR pin up to make the COM port exit the data mode. To prevent **+++** from being misinterpreted as data, the following sequence should be followed:

- 1) Do not input any character for at least 1 second before and after inputting **+++**.
- 2) Input **+++** within 1 second, and wait until **OK** is returned. Once **OK** is returned, the COM port exits the data mode and switches to AT command mode.

If you are exiting the data mode by pulling the MAIN_DTR pin up, make sure to set **AT&D1** first.

● Enter Data Mode

Once **AT+QFTPPUT**, **AT+QFTPGET**, **AT+QFTPLIST**, **AT+QFTPMLSD** and **AT+QFTPNLST** are executed, if the local file path is "**COM:**", which means that data will be received from or outputted to COM port, the COM port enters data mode. If you input **+++** or pull the MAIN_DTR pin up to make the port exit the data mode, in case of **AT+QFTPGET**, **AT+QFTPLIST** and **AT+QFTPNLST** the COM port re-enters data mode after you execute **ATO**, while in case of **AT+QFTPPUT** the COM port cannot re-enter data mode even if you execute **ATO**.

NOTE

See **document [4]** for more information on **AT&D** and **ATO**.

2 Description of FTP(S) AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. AT Commands Description

2.3.1. AT+QFTPCFG Configure Parameters for FTP(S) Server

This command configures FTP(S) server parameters, including username, data type, transfer mode, PDP context ID, response timeout, SSL type, SSL context ID, data connection address selection, etc. If only one parameter is executed, the Write Command queries current settings.

AT+QFTPCFG Configure Parameters for FTP(S) Server	
Test Command AT+QFTPCFG=?	Response +QFTPCFG: "account",<username>,<password> +QFTPCFG: "filetype",(list of supported <file_type>s) +QFTPCFG: "transmode",(list of supported <transmode>s) +QFTPCFG: "contextid",(range of supported <contextID>s) +QFTPCFG: "rsptimeout",(range of supported <timeout>s) +QFTPCFG: "ssltype",(range of supported <SSL_type>s) +QFTPCFG: "sslctxid",(range of supported <SSL_ctx_ID>s) +QFTPCFG: "data_address",(list of supported <data_address>s) OK
Write Command AT+QFTPCFG="account",<username>,<password>]	Response If the optional parameters are omitted, query the current setting: +QFTPCFG: "account",<username>,<password> OK If the optional parameters are specified, set the username and password for authentication: OK Or +CME ERROR: <result>

Write Command AT+QFTPCFG="filetype",<file_type>]	Response If the optional parameter is omitted, query the current setting: +QFTPCFG: "filetype",<file_type> OK If the optional parameter is specified, set the data type: OK Or +CME ERROR: <result>
Write Command AT+QFTPCFG="transmode",<transmode>]*	Response If the optional parameter is omitted, query the current setting: +QFTPCFG: "transmode",<transmode> OK If the optional parameter is specified, set it to the FTP(S) server or a client that listens on a port for data connection: OK Or +CME ERROR: <result>
Write Command AT+QFTPCFG="contextid",<contextID>]	Response If the optional parameter is omitted, query the current setting: +QFTPCFG: "contextid",<contextID> OK If the optional parameter is specified, set the PDP context ID: OK Or +CME ERROR: <result>
Write Command AT+QFTPCFG="rsptimeout",<timeout>]	Response If the optional parameter is omitted, query the current setting: +QFTPCFG: "rsptimeout",<timeout> OK If the optional parameter is specified, set the timeout value for the response of relevant FTP(S) AT commands: OK Or +CME ERROR: <result>
Write Command AT+QFTPCFG="ssltype",<SSL_type>]	Response If the optional parameter is omitted, query the current setting:

e>]	+QFTPCFG: "ssltype",<SSL_type> OK If the optional parameter is specified, set the module to work as an FTP client or FTPS client: OK Or +CME ERROR: <result>
Write Command AT+QFTPCFG="sslctxid",<SSL_ctx_ID>]	Response If the optional parameter is omitted, query the current setting: +QFTPCFG: "sslctxid",<SSL_ctx_ID> OK If the optional parameter is specified, set the SSL context ID: OK Or +CME ERROR: <result>
Write Command AT+QFTPCFG="data_address",<data_address>]	Response If the optional parameter is omitted, query the current setting: +QFTPCFG: "data_address",<data_address> OK If the optional parameter is specified, set the FTP(S) data connection address: OK Or +CME ERROR: <result>

Parameter

<username>	String type. Username for authentication. Maximum parameter size: 255 bytes.
<password>	String type. Password for authentication. Maximum parameter size: 255 bytes.
<file_type>	Integer type. Type of transferred data. <u>0</u> Binary 1 ASCII
<transmode>	Integer type. Determines whether the FTP(S) server or a client listens on a port for data connection. <u>0</u> Active mode, the module will listen on a port for data connection <u>1</u> Passive mode, the FTP(S) server will listen on a port for data connection
<contextID>	Integer type. PDP context ID. Range: 1–5. Default value: 1. It should be activated with AT+QIACT before using AT+QFTPOPEN . For more information on AT+QIACT , see <i>document [1]</i> .

<timeout>	Integer type. Range: 20–180. Default value: 90. Unit: second. Generally, it is the timeout value for the OK result code that is returned for most FTP(S) AT commands, except AT+QFTPPUT , AT+QFTPGET , AT+QFTPLST , AT+QFTPMLSD and AT+QFTPNLST . The rules for these four commands are shown below: a) This parameter indicates the maximum timeout value for CONNECT that is outputted after the command has been sent. b) When the module has entered the data mode, this parameter indicates the maximum interval between two packets of received/transmitted data. c) When <local_name> is not "COM:" , this parameter indicates the maximum interval between two packets of received/transmitted data.
<SSL_type>	Integer type. Determines whether the module works as an FTP client or an FTPS client. 0 FTP client 1 FTPS client (FTP over implicit TLS/SSL) 2 FTPS client (FTP over explicit TLS/SSL)
<SSL_ctx_ID>	Integer type. SSL context ID. Range: 0–5. Default value: 0. Configure the SSL parameters by executing AT+QSSLCFG . For more information, see document [2] .
<data_address>	Integer type. Determines the FTP(S) data connection address. 0 Use server-dispatched address 1 Use FTP(S) control session address
<result>	Integer type. Result code. See Chapter 5 for more information.

NOTE

During FTPS operation, **<transmode>** must be set to 1 because FTPS does not support active mode currently.

2.3.2. AT+QFTPOPEN Login to FTP(S) Server

This command logs in the FTP(S) server. If **OK** is returned, **+QFTPOPEN: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**.

AT+QFTPOPEN Login to FTP(S) Server

Test Command AT+QFTPOPEN=?	Response +QFTPOPEN: <hostname>,<port> OK
Write Command AT+QFTPOPEN=<hostname>[,<port>]	Response OK +QFTPOPEN: <result>,<protocol_error> If there is any error:

+CME ERROR: <result>

Parameter

<hostname>	String type. IP address or domain name of FTP(S) server. Maximum parameter size: 200 bytes.
<port>	Integer type. FTP(S) server port number. Default value: 21.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in the FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

NOTE

Port numbers of FTPS and FTP servers are different. FTPS server port number depends on the server provider, and it is usually 990.

2.3.3. AT+QFTPCWD Configure the Current Directory on FTP(S) Server

This command configures the current directory on the FTP(S) server. If **OK** is returned, **+QFTPCWD: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. All file and directory operations will be performed in the current directory.

AT+QFTPCWD Configure the Current Directory on FTP(S) Server

Test Command AT+QFTPCWD=?	Response +QFTPCWD: <path_name> OK
Write Command AT+QFTPCWD=<path_name>	Response OK +QFTPCWD: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<path_name>	String type. Directory path on FTP(S) server. Maximum parameter size: 255 bytes. FTP(S) server root path: "/" .
<result>	Integer type. Result code. See Chapter 5 for more information.

<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in the FTP(S) protocol. For more information, see Chapter 5 . If it is 0, it is invalid.
-------------------------------	--

2.3.4. AT+QFTPPWD Get the Current Directory on FTP(S) Server

This command retrieves the current directory on the FTP(S) server. If **OK** is returned, **+QFTPPWD: 0,<path_name>** or **+QFTPPWD: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**.

AT+QFTPPWD Get the Current Directory on FTP(S) Server	
Test Command AT+QFTPPWD=?	Response OK
Execution Command AT+QFTPPWD	Response OK If the current directory is successfully retrieved: +QFTPPWD: 0,<path_name> If the retrieval of current directory fails: +QFTPPWD: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<path_name>	String type. Directory path on FTP(S) server. Maximum parameter size: 255 bytes. FTP(S) server root path: "/" .
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.5. AT+QFTPPUT Upload a File to FTP(S) Server

This command uploads a file to FTP(S) server. If the file data are uploaded via COM port, then the module will enter data mode. Inputting **+++** will abort file uploading. A local file can be uploaded to the FTP(S) server, but the file must be a UFS file. The file can be uploaded to UFS with **AT+QFUPL**, and then uploaded to FTP(S) server with **AT+QFTPPUT**. After the file is uploaded successfully, the local UFS file can be deleted with **AT+QFDEL**. For more information, see **document [3]**.

A file can be uploaded from a specified file position with **<startpos>**. If **<local_name>** is "COM:", **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If **<local_name>** is not "COM:", **OK** is outputted first, and then **+QFTPPUT: 0,<transferlen>** is outputted after data transfer is completed.

If the module has entered the data mode or **<local_name>** is not "COM:", **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transmitted data.

AT+QFTPPUT Upload a File to FTP(S) Server	
Test Command AT+QFTPPUT=?	Response +QFTPPUT: <file_name>,<local_name>,<startpos>,<uploadlen>,<beof> OK
Write Command When <local_name> is "COM:", i.e., to input data via COM port AT+QFTPPUT=<file_name>,"COM:"[,<startpos>[,<uploadlen>,<beof>]]	Response If the module enters the data mode successfully: CONNECT Then input the data via the COM port: OK If the data are successfully uploaded: +QFTPPUT: 0,<transferlen> If the data upload fails: +QFTPPUT: <result>,<protocol_error> If there is any error: +CME ERROR: <result>
Write Command When <local_name> is not "COM:" AT+QFTPPUT=<file_name>,<local_name>[,<startpos>]	Response OK If the data are successfully uploaded: +QFTPPUT: 0,<transferlen> If the data upload fails : +QFTPPUT: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<file_name>	String type. File name on FTP(S) server. Maximum parameter size: 255 bytes.
<local_name>	String type. Local file name. Maximum parameter size: 80 bytes. If it is "COM:", the file data are inputted via the COM port. If it is not "COM:", the data should be uploaded from UFS to the FTP(S) server. After successful uploading, delete the local UFS file with AT+QFDEL (see document [3] for more information).
<startpos>	Integer type. Start position of the file to be uploaded. Default value: 0. If <uploadlen> and <beof> are specified, <startpos> should be the position where the data continue to be uploaded to the same file.
<uploadlen>	Integer type. Length of the data to be uploaded. Unit: byte. It is valid only when <local_name> is "COM:". When the length of the data uploaded via the COM port reaches <uploadlen> , the module exits the data mode.
<beof>	Integer type. Determines whether it is the last data packet to be uploaded. 0 Not the last data packet. When the data length reaches <uploadlen> , the module exits the data mode and +QFTPPUT: 0,<transferlen> is outputted. In such a case, do not disconnect data connection, as the remaining data need to be uploaded to the same file on FTP(S). 1 The last data packet. When the data length reaches <uploadlen> , the module exits the data mode and data connection can be disconnected. After that, +QFTPPUT: 0,<transferlen> is outputted.
<transferlen>	Integer type. Length of successfully transferred data. Unit: byte.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

Example

```
//Upload a file via the COM port to the FTP(S) server.
AT+QFTPPUT="test.txt","COM:",0           //All data will be saved as test.txt on FTP(S) server.
CONNECT
//Input the file data via the COM port.
+++                                     //Exit the data mode.
OK

+QFTPPUT: 0,1000

//Upload a file via the COM port to the FTP(S) server twice in 1024-byte chunks.
AT+QFTPPUT="test.txt","COM:",0,1024,0   //It is not the last 1024-byte chunk of test.txt.
CONNECT
//Input the file data via the COM port.
OK                                     //Data length reaches 1024 bytes.
```

```

+QFTPPUT: 0,1024
AT+QFTPPUT="test.txt","COM:",1024,1024,1 //It is the last 1024-byte chunk of test.txt.
CONNECT
//Input the file data via the COM port.
OK //Data length reaches 1024 bytes.

+QFTPPUT: 0,1024

//Upload a file from UFS to FTP(S) server.
AT+QFUPL="test1.txt",1000,300,1 //Upload a file to UFS and the file will be saved as test1.txt.
//The expected file size is 1000 bytes. In this example, 300 indicates timeout value and 1 indicates the ACK mode.
//For more information, see document [3].

CONNECT
//Input 1000 bytes of data.
+QFUPL: 1000,707

OK
AT+QFLST=""
+QFLST: "test1.txt",1000

OK
AT+QFTPPUT="test.txt","test1.txt",0 //Upload test1.txt to FTP(S) server and save it there as test.txt.

OK

+QFTPPUT: 0,1000
AT+QFDEL="test1.txt" //Delete the local UFS file.
OK

```

2.3.6. AT+QFTPGET Download a File from FTP(S) Server

This command downloads a file from the FTP(S) server. The file can be outputted to the COM port with **AT+QFTPGET=<file_name>,"COM:"**. The module enters the data mode on receiving data from the server. After the data transfer is completed, the module exits the data mode automatically and outputs **+QFTPGET: 0,<transferlen>**. The file can be saved to UFS with **AT+QFTPGET=<file_name>,<local_name>**. After the file transfer is completed, the module outputs **+QFTPGET: 0,<transferlen>**.

If **<local_name>** is **"COM:"**, **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If **<local_name>** is not **"COM:"**, **OK** is outputted first, and then **+QFTPGET: 0,<transferlen>** after the data transfer is completed.

If the module has entered the data mode or **<local_name>** is not "COM:", **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transmitted data.

AT+QFTPGET Download a File from FTP(S) Server

Test Command AT+QFTPGET=?	Response +QFTPGET: <file_name>,<local_name>,<startpos>,<downloadlen> OK
Write Command When <local_name> is "COM:", i.e., to output the data via COM port AT+QFTPGET=<file_name>,"COM:"[,<startpos>[,<downloadlen>]]	Response If the module enters the data mode successfully: CONNECT <output_data> OK If the data are successfully downloaded: +QFTPGET: 0,<transferlen> If the data download fails: +QFTPGET: <result>,<protocol_error> If there is any error: +CME ERROR: <result>
Write Command When <local_name> is not "COM:" AT+QFTPGET=<file_name>,<local_name>[,<startpos>]	Response OK If the data are successfully downloaded: +QFTPGET: 0,<transferlen> If the data download fails: +QFTPGET: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<file_name>	String type. File name on FTP(S) server. Maximum parameter size: 255 bytes.
<local_name>	String type. Local file name. Maximum parameter size: 80 bytes. If it is "COM:", the file data are outputted to the COM port. If it is not "COM:", the data will be saved to UFS and then the file can be read with AT+QFREAD (see document [3] for more information).
<startpos>	Integer type. Start position of the file to be downloaded. Default value: 0. If

	<downloadlen> is specified, <startpos> should be the position where the data continue to be downloaded from the same file.
<downloadlen>	Integer type. Length of the data to be downloaded. Unit: byte. It is valid only when <local_name> is "COM:". When the length of the data downloaded via the COM port reaches <downloadlen>, the module exits the data mode.
<output_data>	File data outputted to the COM port.
<transferlen>	Integer type. Length of actually transferred data. Unit: byte. If the length is less than <downloadlen>, it means that the file transfer is completed.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

Example

```
//Download a file from FTP(S) server and the data will be outputted to COM port.
AT+QFTPGET="test.txt","COM:",0
CONNECT
<output_data>
OK

+QFTPGET: 0,1000

//Download a file from FTP(S) server and the data will be outputted to COM port twice in 500-byte chunks.
AT+QFTPGET="test.txt","COM:",0,500 //The size of test.txt is 1000 bytes. Download the first
                                     500 bytes.
CONNECT
<output_data>
OK

+QFTPGET: 0,500

AT+QFTPGET="test.txt","COM:",500,500 //Download the remaining 500 bytes.
CONNECT
<output_data>
OK

+QFTPGET: 0,500

//Download a file from FTP(S) server and save it to UFS.
AT+QFTPGET="test.txt","test2.txt",0 //Download the file and save it to UFS as test2.txt.
OK

+QFTPGET: 0,1000
AT+QFLST=""
```

```
+QFLST: "test2.txt",1000
```

```
OK
```

2.3.7. AT+QFTPSIZE Get File Size on FTP(S) Server

The command retrieves the file size on the FTP(S) server. If **OK** is returned, **+QFTPSIZE: 0,<file_size>** or **+QFTPSIZE: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and then reactivated.

AT+QFTPSIZE Get File Size on FTP(S) Server	
Test Command AT+QFTPSIZE=?	Response +QFTPSIZE: <file_name> OK
Write Command AT+QFTPSIZE=<file_name>	Response OK If the file size is successfully retrieved: +QFTPSIZE: 0,<file_size> If the retrieval of file size fails : +QFTPSIZE: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<file_name>	String type. File name on FTP(S) server. Maximum parameter size: 255 bytes.
<file_size>	Integer type. Size of file on FTP(S) server. Unit: byte.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.8. AT+QFTPDEL Delete a File on FTP(S) Server

This command deletes a specified file on the FTP(S) server. If **OK** is returned, **+QFTPDEL: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**.

Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and then reactivated.

AT+QFTPDEL Delete a File on FTP(S) Server

Test Command AT+QFTPDEL=?	Response +QFTPDEL: <file_name> OK
Write Command AT+QFTPDEL=<file_name>	Response OK +QFTPDEL: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<file_name>	String type. File name on FTP(S) server. Maximum parameter size: 255 bytes.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.9. AT+QFTPMKDIR Create a Folder on FTP(S) Server

This command creates a folder on the FTP(S) server. If **OK** is returned, **+QFTPMKDIR: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and then reactivated.

AT+QFTPMKDIR Create a Folder on FTP(S) Server

Test Command AT+QFTPMKDIR=?	Response +QFTPMKDIR: <folder_name> OK
Write Command AT+QFTPMKDIR=<folder_name>	Response OK +QFTPMKDIR: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<folder_name>	String type. Folder name on FTP(S) server. Maximum parameter size: 255 bytes.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.10. AT+QFTPRMDIR Delete a Folder on FTP(S) Server

This command deletes a specified folder on the FTP(S) server. If **OK** is returned, **+QFTPRMDIR: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and then reactivated.

AT+QFTPRMDIR Delete a Folder on FTP(S) Server	
Test Command AT+QFTPRMDIR=?	Response +QFTPRMDIR: <folder_name> OK
Write Command AT+QFTPRMDIR=<folder_name>	Response OK +QFTPRMDIR: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<folder_name>	String type. Folder name on FTP(S) server. Maximum parameter size: 255 bytes.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.11. AT+QFTPLIST List Content of a Directory on FTP(S) Server

This command lists the content of a directory on the FTP(S) server. If **<local_name>** is **"COM:"**, **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If **<local_name>** is not **"COM:"**, **OK** is returned first, and then **+QFTPLIST: 0,<transfer_size>** is outputted after the content transfer is completed.

If the module has entered the data mode or **<local_name>** is not "COM:", **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transmitted data.

AT+QFTPLIST List Content of a Directory on FTP(S) Server

Test Command AT+QFTPLIST=?	Response +QFTPLIST: <dirname>,<local_name> OK
Write Command When <local_name> is "COM:" AT+QFTPLIST= <dirname>[,"COM:"]	Response If the module enters the data mode successfully: CONNECT <output_content> OK If the content of the directory is successfully listed: +QFTPLIST: 0,<transfer_size> If the content listing fails: +QFTPLIST: <result>,<protocol_error> If there is any error: +CME ERROR: <result>
Write Command When <local_name> is not "COM:" AT+QFTPLIST=<dirname>,<local_name>	Response OK If the content of the directory is successfully listed: +QFTPLIST: 0,<transfer_size> If the content listing fails: +QFTPLIST: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<dirname>	String type. Directory name on FTP(S) server. Maximum parameter size: 255 bytes. If it is ".", the command will list the content of the current directory configured with AT+QFTPCWD .
<local_name>	String type. Local storage location of the data from the FTP(S) server. Maximum parameter size: 80 bytes. Default value: "COM:". If it is "COM:", the data will be outputted to COM port. If it is not "COM:", the data will be saved to UFS and then can be read with AT+QFREAD (see document [3] for more information).

<output_content>	The directory content outputted to the COM port.
<transfer_size>	Integer type. Size of data transferred from the FTP(S) server. Unit: byte.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

Example

```
//List the content of the current directory on the FTP(S) server and the data will be outputted to the COM
port.
AT+QFTPLIST="."
CONNECT
<output_content>
OK

+QFTPLIST: 0,1000

//List the content of a specified directory on the FTP(S) server and save it to UFS.
AT+QFTPLIST="TESTDIR","test2.txt"
OK

+QFTPLIST: 0,1000
AT+QFLST="*"
+QFLST: "test2.txt",1000

OK
```

2.3.12. AT+QFTPNLST List File Names of a Directory on FTP(S) Server

This command lists file names of a directory on the FTP(S) server. If <local_name> is "COM:", **CONNECT** should be outputted within <timeout> configured with **AT+QFTPCFG**. If <local_name> is not "COM:", **OK** is returned first, and then **+QFTPNLST: 0,<transfer_size>** is outputted after the file name transfer is completed.

If the module has entered the data mode or <local_name> is not "COM:", <timeout> configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transmitted data.

AT+QFTPNLST List File Names of a Directory on FTP(S) Server

Test Command	Response
AT+QFTPNLST=?	+QFTPNLST: <dirname>,<local_name>
	OK

Write Command When <local_name> is "COM:" AT+QFTPNLST=<dirname>[,"COM:"]	Response If the module enters the data mode successfully: CONNECT <output_filename> OK If the file names are successfully listed: +QFTPNLST: 0,<transfer_size> If the file name listing fails: +QFTPNLST: <result>,<protocol_error> If there is any error: +CME ERROR: <result>
Write Command When <local_name> is not "COM:" AT+QFTPNLST=<dirname>,<local_name>	Response OK If the file names of the directory are successfully listed: +QFTPNLST: 0,<transfer_size> If the file name listing fails: +QFTPNLST: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<dirname>	String type. Directory name on FTP(S) server. Maximum parameter size: 255 bytes. If it is ".", the command will list the file names of the current directory configured with AT+QFTPCWD .
<local_name>	String type. Local storage location of the data from the FTP(S) server. Maximum parameter size: 80 bytes. Default value: "COM:". If it is "COM:", the data will be outputted to the COM port. If it is not "COM:", the data will be saved to UFS and then can be read with AT+QFREAD (see <i>document [3]</i> for more information).
<output_filename>	File names outputted to COM port.
<transfer_size>	Integer type. Size of data transferred from the FTP(S) server. Unit: byte.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

Example

```
//List file names of the current directory on the FTP(S) server and the data will be outputted to COM port.
AT+QFTPNLST="."
CONNECT
<output_filename>
OK

+QFTPNLST: 0,1000

//List file names of a specified directory on the FTP(S) server and save them to UFS.
AT+QFTPNLST="TESTDIR","test2.txt"
OK

+QFTPNLST: 0,1000
AT+QFLST="*"
+QFLST: "test2.txt",1000

OK
```

2.3.13. AT+QFTPMLSD List Standardized File and Directory Information

This command lists standardized file and directory information on the FTP(S) server. If **<local_name>** is "COM:", **CONNECT** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. If **<local_name>** is not "COM:", **OK** is returned first and then **+QFTPMLSD: 0,<transfer_size>** is outputted after the information transfer is completed.

If the module has entered the data mode or **<local_name>** is not "COM:", **<timeout>** configured with **AT+QFTPCFG** indicates the maximum interval between two packets of received/transmitted data.

AT+QFTPMLSD List Standardized File and Directory Information	
Test Command AT+QFTPMLSD=?	Response +QFTPMLSD: <dirname>,<local_name> OK
Write Command When <local_name> is "COM:" AT+QFTPMLSD=<dirname>[,"COM:"]	Response If the module enters the data mode successfully: CONNECT <output_info> OK If file and directory information is successfully listed: +QFTPMLSD: 0,<transfer_size>

	If file and directory information listing fails: +QFTPMLSD: <result>,<protocol_error> If there is any error: +CME ERROR: <result>
Write Command When <local_name> is not "COM:" AT+QFTPMLSD=<dirname>,<local_name>	Response OK If file and directory information is successfully listed: +QFTPMLSD: 0,<transfer_size> If file and directory information listing fails: +QFTPMLSD: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<dirname>	String type. Directory name on FTP(S) server. Maximum parameter size: 255 bytes. If it is ".", the command will list standardized file and directory information of the current directory configured with AT+QFTPCWD .
<local_name>	String type. Local storage location of the data from the FTP(S) server. Maximum parameter size: 80 bytes. Default value: "COM:". If it is "COM:", the data will be outputted to the COM port. If it is not "COM:", the data will be saved to UFS and then can be read with AT+QFREAD (see document [3] for more information).
<output_info>	File and directory information outputted to the COM port.
<transfer_size>	Integer type. Size of data transferred from the FTP(S) server. Unit: byte.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

Example

```
//List standardized file and directory information of the current directory on the FTP(S) server and the data
//will be outputted to COM port.
AT+QFTPMLSD="."
CONNECT
<output_info>
OK

+QFTPMLSD: 0,1000
//List standardized file and directory information of a specified directory on the FTP(S) server and save it
```

to UFS.

AT+QFTPMLSD="TESTDIR","test2.txt"

OK

+QFTPMLSD: 0,1000

AT+QFLST="*"

+QFLST: "test2.txt",1000

OK

2.3.14. AT+QFTPMDTM Get the File Modification Time on FTP(S) Server

This command gets the file modification time on the FTP(S) server. If **OK** is returned, **+QFTPMDTM: 0,<modify_time>** or **+QFTPMDTM: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and then reactivated.

AT+QFTPMDTM Get the File Modification Time on FTP(S) Server

Test Command
AT+QFTPMDTM=?

Response
+QFTPMDTM: <file_name>

OK

Write Command
AT+QFTPMDTM=<file_name>

Response
OK

If the file modification time is successfully retrieved:

+QFTPMDTM: 0,<modify_time>

If the retrieval of file modification time fails:

+QFTPMDTM: <result>,<protocol_error>

If there is any error:

+CME ERROR: <result>

Parameter

<file_name>	String type. File name on FTP(S) server. Maximum parameter size: 255 bytes.
<modify_time>	String type. File modification time on FTP(S) server. Its format depends on server. Format: "YYYYMMDDHHMMSS" or "YYYYMMDDHHMMSS.NNN".
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.15. AT+QFTPRENAME Rename a File or Folder on FTP(S) Server

This command renames a file or folder on the FTP(S) server. If **OK** is returned, **+QFTPRENAME: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the FTP(S) connection should be disconnected, and the network should be deactivated and then reactivated.

AT+QFTPRENAME Rename a File or Folder on FTP(S) Server	
Test Command AT+QFTPRENAME=?	Response +QFTPRENAME: <old_name>,<new_name> OK
Write Command AT+QFTPRENAME=<old_name>,<new_name>	Response OK +QFTPRENAME: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<old_name>	String type. Old file name or folder name on FTP(S) server. Maximum parameter size: 255 bytes.
<new_name>	String type. New file name or folder name on FTP(S) server. Maximum parameter size: 255 bytes.
<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

2.3.16. AT+QFTPLEN Get the Length of Transferred Data

This command gets the length of data transferred between the FTP(S) server and client, after executing **AT+QFTPPUT**, **AT+QFTPGET**, **AT+QFTPNLST**, **AT+QFTPMLSD** and **AT+QFTPLIST**.

AT+QFTPLEN Get the Length of Transferred Data	
Test Command AT+QFTPLEN=?	Response OK
Execution Command AT+QFTPLEN	Response OK +QFTPLEN: 0,<transferlen>

If there is any error:
+CME ERROR: <result>

Parameter

<transferlen>	Integer type. Length of data transferred between the FTP(S) server and client. Unit: byte.
<result>	Integer type. Result code. See Chapter 5 for more information.

2.3.17. AT+QFTPSTAT Get the Status of FTP(S) Server

This command gets the status of the FTP(S) server.

AT+QFTPSTAT Get the Status of FTP(S) Server	
Test Command AT+QFTPSTAT=?	Response OK
Execution Command AT+QFTPSTAT	Response OK +QFTPSTAT: 0,<ftpstat> If there is any error: +CME ERROR: <result>

Parameter

<ftpstat>	Integer type. Current status of FTP(S) server. 0 Opening an FTP(S) server 1 FTP(S) server is open and idle 2 Transferring data to FTP(S) server 3 Closing the FTP(S) server 4 FTP(S) server is closed
<result>	Integer type. Result code. See Chapter 5 for more information.

2.3.18. AT+QFTPCLOSE Log Out from FTP(S) Server

This command logs out from the FTP(S) server. If **OK** is returned, **+QFTPCLOSE: <result>,<protocol_error>** should be outputted within **<timeout>** configured with **AT+QFTPCFG**. Otherwise, the network should be deactivated and then reactivated.

AT+QFTPCLOSE Log Out from FTP(S) Server	
Test Command AT+QFTPCLOSE=?	Response OK
Execution Command AT+QFTPCLOSE	Response OK +QFTPCLOSE: <result>,<protocol_error> If there is any error: +CME ERROR: <result>

Parameter

<result>	Integer type. Result code. See Chapter 5 for more information.
<protocol_error>	Integer type. For reference only. Indicates the original error code from FTP(S) server defined in FTP(S) protocol. For more information, see Chapter 6 . If it is 0, it is invalid.

3 Examples

3.1. Login to FTP Server

//Step 1: Configure and activate the PDP context.

AT+QICSGP=1,1,"CMNBIOT","", "",1

//Configure the PDP context 1. China Mobile's NB-IoT network APN: "CMNBIOT".

OK

AT+QIACT=1

//Activate the PDP context 1.

OK

//Activated successfully.

AT+QIACT?

//Query the state of the PDP context.

+QIACT: 1,1,1,"10.7.157.1"

OK

AT+QFTPCFG="contextid",1

//Configure the PDP context ID as 1 (PDP context 1 has been activated).

OK

//Step 2: Configure user account and transfer settings.

AT+QFTPCFG="account","test","test"

//Set a username and a password.

OK

AT+QFTPCFG="filetype",1

//Set the data type as ASCII.

OK

AT+QFTPCFG="transmode",1

//Set the transfer mode as passive mode.

OK

AT+QFTPCFG="rsptimeout",90

//Set a response timeout value.

OK

//Step 3: Login to the FTP server.

AT+QFTPOPEN="hf.quectel.com",21

OK

+QFTPOPEN: 0,0

3.2. Login to FTPS Server

//Step 1: Configure and activate the PDP context.

AT+QICSGP=1,1,"CMNBIOT","",1 //Configure the PDP context 1. The APN is "CMNBIOT", which stands for China Mobile's NB-IoT network.

OK

AT+QIACT=1 //Activate the PDP context 1.

OK //Activated successfully.

AT+QIACT? //Query the state of the PDP context.

+QIACT: 1,1,1, "10.7.157.1"

OK

AT+QFTPCFG="contextid",1 //Configure the PDP context ID as 1 (PDP context 1 has been activated).

OK

//Step 2: Configure user account and transfer settings.

AT+QFTPCFG="account","test","test" //Set a username and a password.

OK

AT+QFTPCFG="filetype",1 //Set the data type as ASCII.

OK

AT+QFTPCFG="transmode",1 //Set the transfer mode as passive mode.

OK

AT+QFTPCFG="rsptimeout",90 //Set response timeout value.

OK

//Step 3: FTPS configuration

AT+QFTPCFG="ssltype",1 //Set SSL type as 1 (the module works as FTPS client).

OK

AT+QFTPCFG="sslctxid",0 //Select SSL context 0.

OK

//Configure the SSL cipher suite type as 0xFFFF, which means it supports all cipher suite types.

AT+QSSLCFG="ciphersuite",0,0xFFFF

OK

//Configure the SSL security level as 0, which means authentication is not needed.

AT+QSSLCFG="secllevel",0,0

OK

//Configure the SSL version as 4, which means all SSL versions.

AT+QSSLCFG="sslversion",0,4

OK

//Step 4: Log into the FTPS server.

AT+QFTPOPEN="hf.quectel.com",990

```
OK
+QFTPOPEN: 0,0
```

3.3. Work with Folders

```
AT+QFTPCWD="/" //Set the current directory.
OK
+QFTPCWD: 0,0
AT+QFTPPWD //Query the current directory.
OK
+QFTPPWD: 0,"/"
AT+QFTPMKDIR="TEST" //Create a folder as TEST on FTP(S) server.
OK
+QFTPMKDIR: 0,0
AT+QFTPRENAME="TEST","TEST_NEW" //Rename the folder as TEST_NEW.
OK
+QFTPRENAME: 0,0
AT+QFTPRMDIR="TEST_NEW" //Delete the folder.
OK
+QFTPRMDIR: 0,0
```

3.4. Work with Files

```
AT+QFTPCWD="/" //Set the current directory.
OK
+QFTPCWD: 0,0
AT+QFTPPWD //Query the current directory.
OK
+QFTPPWD: 0, "/"
AT+QFTPSIZE="test_my1.txt" //Query the size of test_my1.txt on FTP(S) server.
OK
```

```

+QFTPSIZE: 0,1000
AT+QFTPFILENAME="test_my1.txt","test_new.txt" //Rename the file as test_new.txt.
OK

+QFTPFILENAME: 0,0
AT+QFTPMMDTM="test_new.txt" //Get the file modification time of test_new.txt on FTP(S) server.
OK

+QFTPMMDTM: 0, "20140708110039"
AT+QFTPDEL="test_new.txt" //Delete test_new.txt on FTP(S) server.
OK

+QFTPDEL: 0,0

```

3.5. List File Information or File Names of a Directory

```

AT+QFTPCWD="/" //Set the current directory.
OK

+QFTPCWD: 0,0
AT+QFTPLIST=".", "COM:" //List the content of the current directory on the FTP(S) server
and the data will be outputted to the COM port.

CONNECT
<output_content>
OK

+QFTPLIST: 0,1000
AT+QFTPLIST=".", "list.txt" //List the content of the current directory on the FTP(S) server
and save it to UFS as list.txt.

OK

+QFTPLIST: 0,1000
AT+QFTPLIST="TEST_2", "COM:" //List the content of TEST_2 on the FTP(S) server and the data
will be outputted to the COM port.

CONNECT
<output_content>
OK

+QFTPLIST: 0,1000
AT+QFTPNLST=".", "COM:" //List file names of the current directory on the FTP(S) server
and the data will be outputted to the COM port.

CONNECT

```

```

<output_filename>
OK

+QFTPNLST: 0,1000
AT+QFTPNLST=".", "nlst.txt" //List file names of the current directory on the FTP(S) server
                             and save them to UFS as nlst.txt.

OK

+QFTPNLST: 0,1000
AT+QFTPNLST="TEST_2", "COM:" //List file names of TEST_2 on the FTP(S) server and the data
                             will be outputted to the COM port.

CONNECT
<output_filename>
OK

+QFTPNLST: 0,1000
AT+QFTPNLSD=".", "COM:" //List standardized file and directory information of the current
                        directory and the data will be outputted to the COM port.

CONNECT
<output_info>
OK

+QFTPNLSD: 0,1000
AT+QFTPNLSD=".", "nlst.txt" //List standardized file and directory information of the current
                             directory and save it to UFS as nlst.txt.

OK

+QFTPNLSD: 0,1000
AT+QFTPNLSD="TEST_2", "COM:" //List standardized file and directory information of TEST_2 and
                             the data will be outputted to the COM port.

CONNECT
<output_info>
OK

+QFTPNLSD: 0,1000

```

3.6. Upload a File to FTP(S) Server

```

AT+QFTPCWD="/" //Set the current directory.

OK

+QFTPCWD: 0,0

```

```

AT+QFTPSTAT                                     //Get the current status of FTP(S) server.
+QFTPSTAT: 0,1                                     //FTP(S) server is open and idle.

OK

//Upload a file via the COM port to the FTP(S) server.
AT+QFTPPUT="test_my.txt","COM:",0               //All data will be saved as test_my.txt on FTP(S) server.
CONNECT
//Input the file data via the COM port.
+++                                                //Exit the data mode.
OK

+QFTPPUT: 0,1000
AT+QFTPLEN
OK

+QFTPLEN: 0,1000
AT+QFTPSIZE="test_my.txt"
OK

+QFTPSIZE: 0,1000

//Upload a file via the COM port to the FTP(S) server and the start position is 1000.
AT+QFTPPUT="test_my.txt","COM:",1000           //All data will be saved as test_my.txt on FTP(S) server.
CONNECT
//Input the file data via the COM port.
+++                                                //Exit the data mode.
OK

+QFTPPUT: 0,500

AT+QFTPSIZE="test_my.txt"
OK

+QFTPSIZE: 0,1500

//Upload a file via the COM port to FTP(S) server twice in 1024-byte chunks.
AT+QFTPPUT="test.txt","COM:",0,1024,0           //It is not the last 1024-byte chunk of test.txt.
CONNECT
//Input the file data via the COM port.
OK                                                //Data length reaches 1024 bytes.

+QFTPPUT: 0,1024
AT+QFTPPUT="test.txt","COM:",1024,1024,1       //It is the last 1024-byte chunk of test.txt.

```


CONNECT

//Input the file data via the COM port.

OK

//Data length reaches 1024 bytes.

+QFTPPUT: 0,1024

//Upload a file from UFS to FTP(S) server.

AT+QFUPL="test_ufs.txt",1000,300,1

//Upload a file to UFS and the file will be saved as *test_ufs.txt*. The expected file size is 1000 bytes. In this example, 300 indicates timeout value and 1 indicates the ACK mode. For more information, see [document \[3\]](#).

CONNECT

//Input 1000 bytes of data.

+QFUPL: 1000,707

OK

AT+QFLST=""

+QFLST: "test_ufs.txt",1000

OK

AT+QFTPPUT="test_my1.txt","test_ufs.txt",0

//Upload *test_ufs.txt* file to FTP(S) server and save it there as *test_my1.txt*.

OK

+QFTPPUT: 0,1000

AT+QFTPLEN

OK

+QFTPLEN: 0,1000

AT+QFTPSIZE="test_my1.txt"

OK

+QFTPSIZE: 0,1000

AT+QFDEL="test_ufs.txt"

//Delete the local UFS file.

OK

3.7. Download a File from FTP(S) Server

AT+QFTPCWD="/"

//Set the current directory.

OK

```
+QFTPCWD: 0,0
```

```
//Solution 1: Output downloaded data directly via the COM port.
```

```
//Download a file from FTP(S) server and the data will be outputted to COM port.
```

```
AT+QFTPGET="test_my.txt","COM:"
```

```
CONNECT
```

```
<output_data>
```

```
OK
```

```
+QFTPGET: 0,1000
```

```
//Download a file from FTP(S) server and the data will be outputted to COM port twice in 500-byte chunks.
```

```
AT+QFTPGET="test.txt","COM:",0,500 //The size of test.txt file is 1000 bytes. Download the first 500 bytes.
```

```
CONNECT
```

```
<output_data>
```

```
OK
```

```
+QFTPGET: 0,500
```

```
AT+QFTPGET="test.txt","COM:",500,500 //Download the remaining 500 bytes.
```

```
CONNECT
```

```
<output_data>
```

```
OK
```

```
+QFTPGET: 0,500
```

```
//Solution 2: Save the downloaded data into a UFS file.
```

```
//Download a file from FTP(S) server and save it to UFS.
```

```
AT+QFTPGET="test_my1.txt","test.txt" //Download the file and save it to UFS as test.txt.
```

```
OK
```

```
+QFTPGET: 0,1000
```

```
AT+QFLST="*"
```

```
+QFLST: test.txt,1000
```

```
OK
```

```
//Download a file from FTP(S) server and save it to UFS. The start position is 450.
```

```
AT+QFTPGET="test_my1.txt","test1.txt",450 //Download the file and save it to UFS as test1.txt.
```

```
OK
```

```
+QFTPGET: 0,550
```

```
AT+QFTPLEN
```

OK

+QFTPLEN: 0,550

AT+QFLST=""

+QFLST: test.txt,1000

+QFLST: test1.txt,550

OK

3.8. Log Out from FTP(S) Server

AT+QFTPCLOSE

//Log out from FTP(S) server.

OK

+QFTPCLOSE: 0,0

AT+QIDEACT=1

//Deactivate the PDP context, which was activated for FTP(S).

OK

4 Error Handling

4.1. FTP(S) AT Command Execution Failure

If **ERROR** response is received from the module when executing FTP(S) AT commands, check whether:

1. the (U)SIM card is inserted
2. **+CPIN: READY** is returned after executing **AT+CPIN?**

4.2. PDP Activation Failure

If the PDP context activation with **AT+QIACT** fails, check the following configurations:

1. Query the PS domain status by executing **AT+CEREG?** to make sure that the PS domain is registered.
2. Query the PDP context parameters by executing **AT+QICSGP=<contextID>** to make sure that the APN of the specified PDP context is set.
3. Make sure that the specified PDP context ID is neither used by PPP nor activated with **AT+CGACT**.
4. The module supports maximum three simultaneously activated PDP contexts under Cat M and maximum two under Cat NB2.

If all above configurations are correct but if the PDP context activation with **AT+QIACT** still fails, reboot the module. After rebooting, check the above configurations at least three times at 10-minute intervals to avoid frequent module rebooting.

4.3. DNS Parsing Failure

When executing **AT+QFTPOPEN**, if **+QFTPOPEN: 604,0** is returned, check the following:

1. Make sure that the domain name of the FTP(S) server is valid.
2. Query the status of the PDP context with **AT+QIACT?** to make sure that the specified PDP context has been activated successfully.

4.4. Error Response from FTP(S) Server

When the **<protocol_error>** in **+QFTPXX: <result>,<protocol_error>** is not 0, it indicates an error code from FTP(S) server.

The issue can be identified based on the returned protocol error code. For example, if **<protocol_error>** is 530 (not logged in), **<username>** or **<password>** may be wrong. If **<protocol_error>** is 550 (requested action not taken: file unavailable), the file or directory may not exist. For more information, see RFC 959 - File Transfer Protocol.

5 Summary of Result Codes

The result code **<result>** indicates a result related to mobile equipment or network operation. The detailed description of **<result>** is provided in the table below.

Table 3: Summary of Result Codes

<result>	Description
0	Operation successful
601	Unknown error
602	FTP(S) server blocked
603	FTP(S) server busy
604	DNS parsing failed
605	Network error
606	Control connection closed
607	Data connection closed
608	Socket closed by a peer
609	Timeout error
610	Invalid parameter
611	Failed to open file
612	File position invalid
613	File error
614	Service not available, closing control connection
615	Open data connection failed
616	Connection closed; transfer aborted

617	Requested file action not taken
618	Requested action aborted: local error in processing
619	Requested action not taken: insufficient system storage
620	Syntax error, command unrecognized
621	Syntax error in parameters or arguments
622	Command not implemented
623	Bad sequence of commands
624	Command parameter not implemented
625	Not logged in
626	Account for storing files required
627	Requested action not taken
628	Requested action aborted: page type unknown
629	Requested file action aborted
630	Requested file name invalid
631	SSL authentication failed

6 Summary of FTP(S) Error Codes

The protocol error code **<protocol_error>** indicates an error response from FTP(S) server when it is not 0. For more information, see *RFC 959 - File Transfer Protocol*. The detailed description of **<protocol_error>** is provided in the table below.

Table 4: Summary of FTP(S) Error Codes

<protocol_error>	Description
421	Service not available, closing control connection
425	Opening data connection failed
426	Connection closed; transfer aborted
450	Requested file action not taken
451	Requested action aborted: local error in processing
452	Requested action not taken: insufficient system storage
500	Syntax error, command unrecognized
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command parameter not implemented
530	Not logged in
532	Need account for storing files
550	Requested action not taken: file unavailable
551	Requested action aborted: page type unknown
552	Requested file action aborted: exceeded storage allocation
553	Requested action not taken: file name not allowed

7 Appendix References

Table 5: Related Documents

Document Name
[1] Quectel_BG770A-GL&BG95xA-GL_TCP(IP)_Application_Note
[2] Quectel_BG770A-GL&BG95xA-GL_SSL_Application_Note
[3] Quectel_BG770A-GL&BG95xA-GL_FILE_Application_Note
[4] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual

Table 6: Terms and Abbreviations

Abbreviation	Description
ACK	Acknowledgement
APN	Access Point Name
ASCII	American Standard Code for Information Interchange
COM port	Communication port
DNS	Domain Name Server
DTR	Data Terminal Ready
FTP	File Transfer Protocol
FTPS	FTP over SSL
ID	Identifier
IP	Internet Protocol
LTE	Long-Term Evolution

NB	Narrowband
NB-IoT	Narrowband Internet of Things
PDP	Packet Data Protocol
PPP	Point-to-Point Protocol
PS	Packet Switching
SSL	Secure Sockets Layer
TLS	Transport Layer Security
UFS	User File Storage
URC	Unsolicited Result Code
(U)SIM	(Universal) Subscriber Identity Module
